

# Financial Econometrics - Homework 6

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## Chapter 10: Random Regressors and Moment-Based Estimation

Name: [Your Name]

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### Question 10.18

**a. Create MOTHERCOLL and FATHERCOLL. What percentage of parents have some college education?**

Percentage of mothers with some college education ( $\text{MOTHEREDUC} > 12$ ): 12.15%

Percentage of fathers with some college education ( $\text{FATHEREDUC} > 12$ ): 11.68%

**b. Correlations between EDUC, MOTHERCOLL, and FATHERCOLL. Logical argument.**

Correlations:

-  $\text{Corr}(\text{EDUC}, \text{MOTHERCOLL}) = 0.3595$

-  $\text{Corr}(\text{EDUC}, \text{FATHERCOLL}) = 0.3985$

-  $\text{Corr}(\text{MOTHERCOLL}, \text{FATHERCOLL}) = 0.3546$

Argument: MOTHERCOLL and FATHERCOLL are indicators of higher education. They might be better instruments than the continuous years of education because they are less prone to measurement error and may more directly reflect the "threshold" of socioeconomic status that influences the daughter's education while being potentially less correlated with the daughter's innate ability.

**c. Estimate the wage equation using MOTHERCOLL as an IV. 95% interval estimate for the coefficient of EDUC.**

The IV estimate for the EDUC coefficient has a 95% confidence interval of  $[-0.0012, 0.1533]$ .

**d. Estimate the first-stage equation and test the strength of MOTHERCOLL.**

First-stage F-statistic for MOTHERCOLL = 0:  $F = 63.563$ ,  $p\text{-value} < 2.2e-16$ .

Conclusion: Since  $F > 10$ , MOTHERCOLL is a strong instrumental variable.

**e. Estimate the wage equation using both MOTHERCOLL and FATHERCOLL as IVs.**

The IV estimate for the EDUC coefficient has a 95% confidence interval of [0.0275, 0.1482].

Comparison: This interval is narrower than the one in part (c), indicating that using more instruments (and likely a stronger combined instrument) increases the precision of the estimate.

**f. Estimate the first-stage equation for part (e) and test joint significance.**

First-stage F-statistic for MOTHERCOLL = 0 and FATHERCOLL = 0:  $F = 56.963$ ,  $p\text{-value} < 2.2\text{e-}16$ .

Conclusion: The instruments are adequately strong.

**g. Test the validity of the surplus instrument (Sargan test).**

Sargan test statistic: 0.238,  $p\text{-value} = 0.626$ .

Conclusion: Since  $p > 0.05$ , we fail to reject the null hypothesis. The surplus instrument is valid.

## Question 10.20

**a. OLS estimate of CAPM for Microsoft.**

Coefficient (Beta): 1.2018, SE: 0.1221,  $t\text{-value} = 9.839$ ,  $p\text{-value} < 2.2\text{e-}16$ .

Classification: Microsoft stock is "risky" since its beta (1.2018) is greater than 1, meaning it is more volatile than the market portfolio.

**b. Rank-based IV. First-stage regression results.**

First-stage regression (Market Return on RANK):

- RANK Coefficient:  $9.067\text{e-}04$ ,  $t\text{-value} = 43.1$ ,  $p\text{-value} < 2\text{e-}16$ .

- R-squared: 0.9126.

Conclusion: RANK is a very strong instrumental variable.

**c. Hausman test for exogeneity of market return.**

Augmented regression coefficient on first-stage residuals ( $\hat{v}$ ): -0.8746,  $t\text{-value} = -2.040$ ,  $p\text{-value} = 0.0428$ .

Conclusion: At the 5% significance level, we reject the null hypothesis of exogeneity ( $p < 0.05$ ). Market return is likely endogenous.

**d. IV/2SLS estimate using RANK.**

IV Estimate (Beta): 1.2783, SE: 0.1280.

Comparison: The IV estimate (1.2783) is slightly higher than the OLS estimate (1.2018), which is consistent with the presence of measurement error causing attenuation bias in OLS.

**e. RANK and POS as IVs. First-stage results.**

Joint F-test for RANK and POS in first-stage:  $F = 951.26$ ,  $p\text{-value} < 2.2e-16$ .

Conclusion: The instruments are adequately strong.

**f. Hausman test for part (e).**

Coefficient on first-stage residuals ( $\hat{v}_e$ ): -0.9549,  $t\text{-value} = -2.205$ ,  $p\text{-value} = 0.0287$ .

Conclusion: We reject exogeneity at the 5% level.

**g. IV/2SLS estimates using RANK and POS.**

IV Estimate (Beta): 1.2831, SE: 0.1279.

Comparison: This estimate is very close to the single-IV estimate in part (d).

**h. Sargan test for the validity of surplus IV.**

Sargan test statistic: 0.5585,  $p\text{-value} = 0.4549$ .

Conclusion: We fail to reject the null hypothesis. The surplus instrument is valid.